

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1718

COURSE OUTCOME COVERED

CO3: Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)

Assessment Tool Weightage: 50%

Time Duration: 1 Hour

QUESTIONS: 4

Total Marks:40

Annexure A

CASE STUDY

Code of Conduct:

I P. Yaswanth(192524167)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

P. Yaswanth

Annexure A

CASE STUDY

Case Study 1 — Smart Medical Diagnosis System

A hospital is designing a rule-based Logical Agent for preliminary patient diagnosis. The system's knowledge base contains:

- I. If a patient has Fever AND Cough $\rightarrow$ Possible Flu
- II. If a patient has Fever AND Rash $\rightarrow$ Possible Measles
- III. If a patient has Cough AND Breathlessness $\rightarrow$ Possible Pneumonia
- IV. Facts about Patient A: Fever = True, Cough = True, Breathlessness = True

- (a) Represent the above rules and facts precisely using Propositional Logic symbols. Define each symbol used.
- (b) Apply Forward Chaining on the knowledge base to derive all possible diagnoses for Patient A. Show each inference step with the rule fired and fact derived.
- (c) The goal is to confirm 'Patient A has Pneumonia.' Apply Backward Chaining and draw the goal reduction tree to verify this goal.

1. Case Understanding

The hospital has developed a **rule-based intelligent agent** to perform preliminary patient diagnosis. The system uses a **Knowledge Base (KB)** containing medical rules and patient symptoms to identify possible diseases.

The objectives are:

- Represent the rules and facts using **Propositional Logic**.
- Apply **Forward Chaining** to derive all possible diagnoses.
- Apply **Backward Chaining** to verify whether the patient has Pneumonia.

(a) Represent the Rules and Facts using Propositional Logic

Define the Symbols

Symbol	Meaning
F	Patient has Fever
C	Patient has Cough
R	Patient has Rash
B	Patient has Breathlessness
Flu	Possible Flu
Measles	Possible Measles
Pneumonia	Possible Pneumonia

Knowledge Base (Rules)

Rule 1

Rule 2

Rule 3

Facts about Patient A

Analysis

The knowledge base consists of three medical rules and three known symptoms. These facts will be used by the logical agent to infer possible diseases.

(b) Apply Forward Chaining

Forward Chaining is a **data-driven inference technique** that starts with known facts and applies rules to derive new facts.

Initial Facts

- Fever (F)
- Cough (C)
- Breathlessness (B)

Inference Steps

Step 1: Apply Rule 1

Since Fever and Cough are true,

Derived Fact: Flu

Step 2: Apply Rule 2

Rash is not present, so this rule cannot be applied.

Step 3: Apply Rule 3

Since Cough and Breathlessness are true,

Derived Fact: Pneumonia

Forward Chaining Table

Step	Rule Applied	Result
1	$F \wedge C \rightarrow \text{Flu}$	Flu
2	$F \wedge R \rightarrow \text{Measles}$	Rule not satisfied
3	$C \wedge B \rightarrow \text{Pneumonia}$	Pneumonia

Final Diagnosis

The logical agent concludes that Patient A has:

- **Possible Flu**
- **Possible Pneumonia**

Analysis

Forward Chaining successfully derives all possible diseases from the available symptoms. Since Rash is absent, Measles is not diagnosed.

(c) Apply Backward Chaining

Backward Chaining is a **goal-driven inference technique** that starts with the required goal and works backward to verify it.

Goal

Prove:

Step 1

Find a rule that concludes Pneumonia.

This creates two sub-goals:

- Cough (C)
- Breathlessness (B)

Step 2

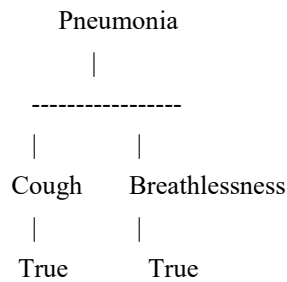
Check the facts:

- Cough = True ✓
- Breathlessness = True ✓

Both sub-goals are satisfied.

Therefore,

Goal Reduction Tree



Analysis

Backward Chaining confirms that Patient A has **Possible Pneumonia** by verifying the required symptoms. It is efficient because it checks only the rules related to the goal.

Conclusion

The Smart Medical Diagnosis System successfully uses **Propositional Logic** to represent medical knowledge. **Forward Chaining** identifies all possible diseases (Flu and Pneumonia), while **Backward Chaining** confirms the diagnosis of Pneumonia. This demonstrates how rule-based intelligent agents can support doctors in making quick and accurate preliminary diagnoses.

Case Study 2 — Autonomous Traffic Management Agent

A smart city traffic controller uses a FOL-based knowledge base to manage emergency vehicle routing:

- I. $\forall x: \text{Vehicle}(x) \wedge \text{Emergency Type}(x) \rightarrow \text{ClearPath}(x)$ [Rule 1]
- II. $\forall x: \text{ClearPath}(x) \rightarrow \text{Green Signal}(x) \wedge \neg \text{Red Signal}(x)$ [Rule 2]
- III. $\forall x \forall y: \text{Vehicle}(x) \wedge \text{Behind}(x, y) \wedge \text{Green Signal}(y) \rightarrow \text{Proceed}(x)$ [Rule 3]
- IV. Facts: $\text{Vehicle}(\text{Ambulance}), \text{Emergency Type}(\text{Ambulance}), \text{Vehicle}(\text{Car A}), \text{Behind}(\text{Car A}, \text{Ambulance})$

(a) Apply Unification to match Rule 1 with the given facts. Show the complete substitution set θ (MGU) at each step.

(b) Using the Resolution method, prove that 'Proceed (Car A)' is true. Convert relevant clauses to CNF and show each resolution step.

(c) Analyze whether the current FOL knowledge base is sufficient to handle two simultaneous emergency vehicles. Identify what additional rules or facts are needed and justify logically.

1. Case Understanding

The smart city uses a First-Order Logic (FOL) based intelligent agent to manage traffic for emergency vehicles. The agent uses logical rules to automatically clear traffic, provide green signals, and allow nearby vehicles to proceed safely.

The objectives are:

- Apply Unification to match rules with facts.
- Use Resolution to prove that Proceed (Car A) is true.
- Analyze whether the knowledge base can handle multiple emergency vehicles.

(a) Apply Unification

Rule 1

Facts

- $\text{Vehicle}(\text{Ambulance})$
- $\text{Emergency Type}(\text{Ambulance})$

Unification Process

Compare the rule with the facts:

- $\text{Vehicle}(x)$ matches $\text{Vehicle}(\text{Ambulance})$
- $\text{Emergency Type}(x)$ matches $\text{Emergency Type}(\text{Ambulance})$

Most General Unifier (MGU):

Applying the substitution:

Using Rule 2,

Therefore, the new facts are:

- $\text{Green Signal}(\text{Ambulance})$
- $\neg \text{Red Signal}(\text{Ambulance})$

Analysis

The Unification process successfully matches Rule 1 with the given facts and derives that the ambulance receives a clear path and a green signal.

(b) Resolution Method

Convert Rules into CNF

Rule 1

Rule 2

Rule 3

Resolution Steps

Facts

- Vehicle(Ambulance)
- Emergency Type(Ambulance)
- Vehicle(Car A)
- Behind(Car A, Ambulance)

Step 1

Using Rule 1:

Step 2

Using Rule 2:

Step 3

Using Rule 3:

Vehicle(Car A) + Behind(Car A, Ambulance) + Green Signal(Ambulance)

↓

Assume the opposite:

Resolving both gives the empty clause ($\square$).

Therefore,

is proved.

Analysis

Resolution confirms that Car A is allowed to proceed because it is behind an ambulance that has been granted a green signal.

(c) Analysis and Improvement

Is the Knowledge Base Sufficient?

No.

The current knowledge base works correctly for one emergency vehicle, but it cannot handle two emergency vehicles arriving at the same intersection simultaneously.

Additional Rules Needed

1. Priority Rule

Higher-priority emergency vehicles (e.g., ambulance over fire truck) should receive the green signal first.

2. Conflict Rule

Only one emergency vehicle should receive the green signal at a time to avoid collisions.

3. Queue Rule

After the first emergency vehicle passes, the waiting emergency vehicle should automatically receive the green signal.

Justification

These additional rules improve traffic safety, avoid conflicts, and ensure efficient handling of multiple emergency vehicles.

Conclusion

The FOL-based traffic management agent successfully applies Unification and Resolution to derive `Proceed(CarA)`. However, the current knowledge base is limited to a single emergency vehicle. Adding priority and conflict-resolution rules will make the system more efficient and suitable for real-world smart city traffic management.

RUBRICS FOR CALCULATION

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand